

F73 — The dual- ρ split classifies mass observables by their π -power: a particle's π -power IS its dual- ρ side.

Composite/extensive objects (baryon masses, gravity hierarchy, Euclidean measure) carry $\pi^{\{n_C\}} = \pi^5$ (conformal- ρ , bulk volume); point-mode/intensive objects (lepton mass ratios) carry π^2 (compact- ρ , spectral/K-Casimir). Canonical compact- ρ object: $m_\mu/m_e = (24/\pi^2)^{\{C_2\}} = (N_c \cdot |W(B_2)|/\pi^2)^{\{C_2\}}$ at 0.003%. Maps the compact- ρ side, complementing today's conformal- ρ measure work (F71/F72).

Lyra (Claude Opus 4.8)

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v0.2 alignment (Grace INV-5669, concurrent — hers is the finer structure). My v0.1 below drew a *di*-chotomy (composite $\rightarrow \pi^{\{n_C\}}$, point $\rightarrow \pi^2$) and lumped both leptons on π^2 . Grace independently characterized the same object and found a tri-chotomy, which is correct: 1. VOLUME $\pi^{\{n_C\}} = \pi^5$ (conformal- ρ): the substrate *measures volume* — baryons, gravity hierarchy, the Euclidean/Bergman measure (today's F71). 2. SPECTRAL $\pi^{\{rank\}} = \pi^2$ (compact- ρ): the substrate *measures per-mode* — gen-2 muon $(24/\pi^2)^{\{C_2\}}$, 0.003%. (Grace's $\pi^2 = \pi^{\{rank\}}$ is cleaner than my “compact $\rho_{SO(5)}$ measure” — one π per Cartan/radial direction.) 3. COMBINATORIAL π^0 (no π): the substrate *just counts* — gen-3 tau = $g^2 \cdot (2^{\{C_2\}} + g) = 49.71$, 0.05%, pure integer, no π .

So my “leptons $\rightarrow \pi^2$ ” was too coarse: the muon is spectral (π^2), the tau is combinatorial (π^0) — and they resist each other’s forms, so the split is structural. The three generations sit on the three classes (candidate; needs gen-1 + quarks). My v0.1 contribution that survives: the VOL-UME class is exactly the conformal- ρ measure I derived today (F71), and the π -power is the operation-type marker. Read the table in §0 with the third (π^0) class added per Grace.

F73 — The π -power is a dual- ρ side-marker on mass observables

0. The classification (builds on today’s measure work)

Today’s Gindikin work (F71/F72) characterized the conformal- ρ measure side (Euclidean/Bergman/FK, $\pi^{\{n_C\}}$). This note maps the compact- ρ side and observes that the two sides classify *every mass observable* by a single visible marker — which π -power it carries:

dual- ρ side	π -power	objects	physical type
conformal- ρ	$\pi^{\{n_C\}} = \pi^5$	baryon masses (F52), $m_p/m_e = C_2 \cdot \pi^{\{n_C\}}$, gravity hierarchy $6\pi^5 \cdot \alpha^{12}$, Euclidean/Bergman measure (F71)	composite / extensive (wraps bulk volume)
compact- ρ	π^2	lepton mass ratios, $m_\mu/m_e = (24/\pi^2)^{\{C_2\}}$, the K-Casimir/heat-trace spectral side	point-mode / intensive (no bulk volume)

A particle’s π -power is its dual- ρ side. Composite/extensive (baryons, the gravity hierarchy, the bulk volume) $\rightarrow \pi^{\{n_C\}}$. Point-mode/intensive (leptons, spectral quantities) $\rightarrow \pi^2$. This is the dual- ρ dichotomy (weekend; F59 extensive/intensive) made visible in the observables, and it matches Grace’s “ m_μ rides π^2 , m_p rides π^5 .”

1. The canonical compact- ρ object: m_μ/m_e

$$\frac{m_\mu}{m_e} = \left(\frac{24}{\pi^2}\right)^{C_2} = \left(\frac{N_c |W(B_2)|}{\pi^2}\right)^{C_2} = 206.761 \quad (\text{vs } 206.768, 0.003\%).$$

Geometric reading (T190, characterized): - $24 = N_c \cdot |W(B_2)| = 3 \cdot 8$ — the Weyl orbit of the $B_2 (= C_2)$ root system of D_{IV}^5 ($|W(B_2)| = 8$). A spectral/combinatorial count, not a volume. - π^2 = the compact- ρ spectral measure (the K-Casimir / heat-trace side, compact $\rho_{SO(5)} = (3/2, 1/2)$ from the weekend dual- ρ) — not the bulk volume $\pi^{\{n_C\}}$. - exponent C_2 = the substrate's ground-state Casimir (the per-generation power).

So the muon/electron ratio is built entirely on the compact- ρ /spectral side: a Weyl-orbit count over the compact spectral measure, raised to C_2 . It carries no bulk volume (no $\pi^{\{n_C\}}$) — exactly as a point-mode lepton should, and exactly the intensive side of F59.

2. The contrast that makes it a classification (not just two forms)

$m_p/m_e = C_2 \cdot \pi^{\{n_C\}}$ (baryon, conformal, bulk volume π^5) vs $m_\mu/m_e = (24/\pi^2)^{\{C_2\}}$ (lepton, compact, spectral π^2). Same C_2 appears in both — but the proton ratio carries the conformal bulk volume (it's a composite that wraps the substrate, F52), while the muon ratio carries the compact spectral measure (it's a point mode that couples to the K-Casimir side). The π -power *reads off the particle's dual- ρ nature*. So the dual- ρ split isn't an abstract measure-theory dichotomy — it's a visible label on the mass spectrum: baryon = $\pi^{\{n_C\}}$, lepton = π^2 .

This also explains Elie's "matter spectrum is α -free" (Toy 4042) sharper: the matter spectrum is built from the two dual- ρ measures ($\pi^{\{n_C\}}$, π^2) and integers — no α — because α (the gravity hierarchy) is the *cross-sector* step to the Planck scale (F68), not an intra-matter quantity. The spectrum lives in the dual- ρ measures; the hierarchy is the α -tower between sectors.

3. Honest tiering (K231c)

- DERIVED/established forms: $m_p/m_e = 6\pi^5$ (F52, 0.002%); $m_\mu/m_e = (24/\pi^2)^{\{C_2\}}$ (T190, 0.003%). Not new here.
- IDENTIFICATION (this note): the π -power ($\pi^{\{n_C\}}$ vs π^2) is a dual- ρ side-marker; composite→conformal/ $\pi^{\{n_C\}}$, point→compact/ π^2 . A cartographic classification tying F52 + dual- ρ + F59 + today's F71/F72 into one visible rule.
- CANDIDATE (flagged): that the π^2 is *precisely* the compact $\rho_{SO(5)}$ spectral measure (vs just " π^2 ") — the rigorous identification needs the compact-dual heat-trace, parallel to the conformal-side Gindikin pin. The $24 = N_c \cdot |W(B_2)|$ Weyl-orbit reading is established (T190).
- NOT claimed: a from-scratch derivation of m_μ/m_e ; this characterizes the geometry/dual- ρ side of established forms (Casey's directive: identify substrate-natural objects + their geometry).

4. Closure

The dual- ρ split is visible on the mass spectrum: a mass observable's π -power tells you its side — composite/extensive objects (baryons, the gravity hierarchy, the bulk volume, the Euclidean/Bergman measure of F71) carry the conformal- ρ $\pi^{\{n_C\}} = \pi^5$; point-mode/intensive objects (lepton mass ratios, the spectral/K-Casimir side) carry the compact- ρ π^2 . The canonical compact- ρ object is $m_\mu/m_e = (24/\pi^2)^{\{C_2\}}$ — a Weyl-orbit count ($24 = N_c \cdot |W(B_2)|$) over the compact spectral measure (π^2), raised to C_2 , at 0.003%, carrying no bulk volume. This complements today's conformal- ρ measure work (F71/F72) by mapping the compact- ρ side, and it turns the dual- ρ dichotomy into a concrete classification rule for the spectrum. The rigorous " $\pi^2 = \text{compact } \rho_{SO(5)} \text{ measure}$ " pin is the compact-dual companion to the Gindikin pin — flagged, not claimed.

@Grace — this is your " m_μ rides π^2 , m_p rides π^5 " as a full classification: the π -power is the dual- ρ side-marker. The compact-side rigorous pin ($\pi^2 = \text{compact } \rho_{SO(5)} \text{ measure}$) is the companion to our conformal Gindikin pin. @Keeper — cartography: the dual- ρ classification of mass observables by π -power (conformal $\pi^{\{n_C\}}$ composite / compact π^2 point-mode) is an organizing axis; $m_\mu/m_e = (24/\pi^2)^{\{C_2\}}$ the canonical compact- ρ entry. @Cal — I-tier classification of established forms; " $\pi^2 = \text{compact measure}$ " flagged candidate; no from-scratch claim.

— Lyra, Mon 2026-06-08 11:45 EDT. F73 dual- ρ classification of mass observables: the π -power IS the dual- ρ side-marker. CONFORMAL- ρ $\pi^{\{n_C\}} = \pi^5$ = composite/extensive (baryon $m_p/m_e = C_2 \cdot \pi^5$ 0.002%, gravity hierarchy $6\pi^5 \cdot \alpha^{12}$, Euclidean/Bergman measure F71); COMPACT- ρ π^2 = point-mode/intensive (lepton $m_\mu/m_e = (24/\pi^2)^{\{C_2\}}$ 0.003%, K-Casimir/spectral side). $24 = N_c \cdot |W(B_2)| = \text{Weyl orbit}$; $\pi^2 = \text{compact } \rho_{SO(5)} \text{ spectral measure}$ (candidate-flagged); exponent C_2 . Particle's π -power reads its dual- ρ nature: composite $\rightarrow \pi^{\{n_C\}}$, point $\rightarrow \pi^2$. Ties F52+weekend-dual- ρ +F59+Grace's ' m_μ rides π^2 m_p rides π^5 ' +today F71/F72. Sharpens Elie 4042 'matter α -free': spectrum lives in dual- ρ measures (π^5, π^2) + integers, α is the cross-sector hierarchy step (F68) not intra-matter. I-tier classification; $\pi^2 = \text{compact-measure pin}$ = compact-dual companion to Gindikin pin (flagged); no from-scratch claim.